Trustworthy AI Agents Roadmap

Mark Maybury

Lockheed Martin, Chelmsford, MA 01824, USA
mark.t.maybury@lmco.com

Abstract. Rapid adoption of agents across multiple use cases promises productivity, precision, and progress across a range of critical use cases. While Large Language Models (LLMs) and agents promise significant value [1], their increase in capability, agency and autonomy, their impact on individuals and society grow in tandem with individual and organizational risks that require mitigation. This chapter summarizes an interdisciplinary workshop the engaged researchers, practitioners, policy makers and students from diverse fields including computer science, sociology, philosophy, and law to collaboratively create a roadmap considering the scientific and technical, security, economic, environmental, regulatory, social and ethical implications of Agentic AI to advance responsible if not beneficent agents and their enabling ecosystem. This global topic is timely, interdisciplinary, and necessary, addressing a critical need in the AI community and society at large. The chapter concludes by providing an overview of this collection chapter in which address key areas of safety and security and sovereignty, risk measurement and mitigation, and governance.

Keywords: Agents, Safety, Security, Sovereignty, Trust, Roadmap.

1 Introduction

Trustworthy agents must be both safe and secure, which we define as:

- ***Safety***: Design of technology to prevent accidental or unintentional harm. Harms include physical, psychological, cognitive, financial, environmental, reputational, etc. Safety measures can protection from malfunctions, hazards, alignment with human values, training, governance or policing of proper use, controls, guardrails, etc., as revised from ISO/IEC TS 5723:2022 [2].
- **Security**: Avoiding harms by preventing unauthorized use and intentional mis-use of tool or attack on tool by adversary including undermining confidentiality, integrity and availability, as defined in AI RMF 1.0 [3].

Table 1 distinguishes the safety and security of LLMs, agents (which use information and tools including LLMs to accomplish tasks), and robots (physically embodied agents). Where safety aim to protect the world (including humans) from the system, security aims to protect the system from world (often from humans). In the former case we are concerned with unintentional mistakes including errors and accidents whereas in the latter we focus on avoiding intentional or malicious actions. The table entries exemplify safety and security example across LLMs, agents, and robots.

Table 1. Safety and Security of AI: Definitions and Examples

	Safety	Security
Goal	Protect the world from the system	Protect the System from the world
Threat	Unintentional (Errors, Accidents)	Intentional (Malicious)
LLM Example	Misinformation from hallucination or bias	Jailbreaking confidential information with malicious prompts
Agent Example	Revealing personal information without authorization	Manipulating an agent into creating misinformation
Robot Example	Accidentally colliding with a person	Adversary control of a robot

Having distinguished safety and security, we will generalize these into agent risk, and considering both safety and security, define the risk of a system as:

$$\textbf{\textit{Risk}} = \text{Threats (to self/others or by others)} * \text{Vulnerabilities} * \text{Consequences.} \quad (1)$$

A subsequent chapter [4] provides a mathematical model and empirical assessments of leading LLMs using a detailed expansion of this risk measurement formula.

We additionally define sovereign agents:

- ***Sovereign***: Design of technology to ensure local information, operation and control of AI agents via isolation of information (including data and models) or control as required by regulation, policy, or preference typically to ensure safety and security in order to guarantee prevention of accidental, unintentional, or malicious disclosure of information or undesirable action especially in high risk use cases in high assurance domains such as healthcare, finance, defense, or aerospace.

2 Regulations, Standards and Governance

Figure 1 highlights some of the regulatory requirements for safe and secure agents. These vary internationally and locally and range from requirements with harsh fines to suggested practices. For example the EU AI Act [5] is in force with fines for non-compliance whereas the NIST AI Risk Management Framework [3] is a set of best practices for risk reduction. These are complemented by a series of cyber security standards and safety standards. For example, ISO/IEC TS 5723:2022 [2] identifies harms including physical, psychological, cognitive, financial, environmental, reputational, and the need for safety measures that provide protection from malfunctions and hazards, alignment with human values, and training, governance and policing of proper use, controls, and guardrails. Subsequent chapters will explore these and additional aspects of governance in a variety of domains and tasks.

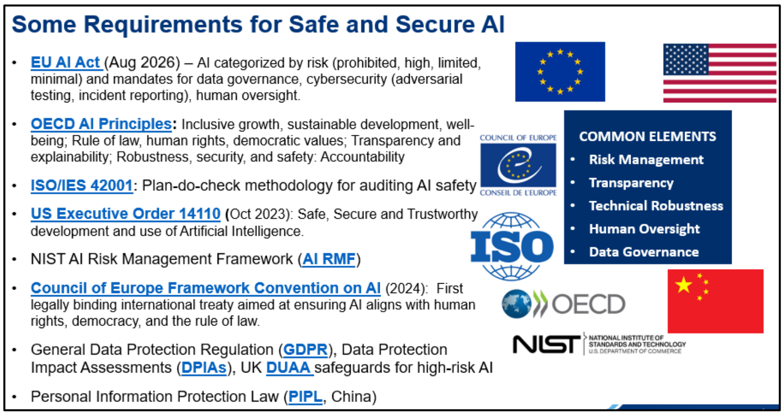

Fig. 1. Regulatory Environment for Safe and Secure AI.

As Figure 2 illustrates, achieving trustworthy agents requires addressing a number of requirements throughout the life cycle, from up front use case through to end of life. First, the agent must be safe and secure. For example, it must assure confidentiality, integrity, and availability vulnerabilities cannot be exploited by adversaries or inadvertent mistakes. This means debiasing, reducing hallucination, and ensuring behavior is aligned with agent design goals and protected with guardrails. In addition, the agent needs to be accurate and transparent meaning validated and verified, a prior and during operation, at provide robust and explainable results. Finally, it much be governed, including real time monitoring and effective human oversight to ensure human trust is appropriate calibrated (avoiding over trust and under trust) to its performance and behavior.

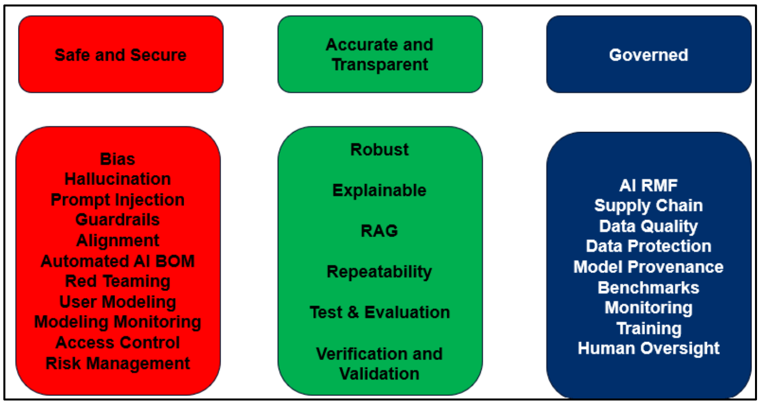

Fig. 2. Trustworthy Agents.

3 Key Questions

When considering agent safety and security, a set of critical questions in key areas include the following.

3.1 Agentic AI

Agentic AI, or goal-seeking AI systems that can act autonomously to achieve specific objectives are expanding in capability and application. While agents promise to revolutionize everything from education to customer service to healthcare, they introduce new challenges such as control (e.g., delegation of authority, mixed-initiative), accountability, and verification and validation of evolving learning systems as well as perils arising from bias, brittleness and belligerence [6] and resultant harms (e.g., financial, physical, social, reputational). How do we ensure and assure progress and avoid unintended and undesirable consequences?

3.2 Models and Autonomy

The global deployment and use of large language/process models, small models for edge devices, and the introduction of large world models is accelerating new cases and advancing agents and autonomous systems across the enterprise and at the edge. Their computational and energy appetite for model training and inference drive a need for resource-efficient infrastructure and processing. How can we accelerate the state of the art in algorithms, architectures, interoperability (e.g., Model Context Protocol) and security to simultaneously advance performance, usability, prosperity, transparency, and safety?

3.3 Society, Safety and Sustainability

Current LLMs suffer from being biased, brittle, and baroque with risk of significant harms. A rapidly expanding number of agents and humanoid robots add new risks of belligerence. Beginning to address these concerns, societies have introduced national guidelines (e.g., NIST AI RMF [3], CAC AI Security Governance Framework [7]), regulations (e.g., DoD 3000.09 [8], Chinese Regulation on the Governance of Generative AI [9]), strategies (e.g., Germany's AI Action Plan [10], US AI Action Plan [11], UK AI Strategy [12]), laws (e.g., EU AI Act [5], GDPR [13]), and sector-specific safety and security guidelines. What are effective methods for responsible development, deployment and operation that can mitigate risks to society? How can governance frameworks, standards, and regulatory mechanisms be effectively implemented and coordinated across jurisdictions and institutions to ensure responsible development and deployment of agentic AI systems?

4 Agent Risk Themes

Several critical sub-themes of agent safety and security include ethics, confidentiality, integrity and availability, human-agent trust, and real world use cases.

4.1 Ethical Foundations and Governance

Ethical foundations and governance address the theoretical and philosophical underpinnings of ethical AI for agents. This encompasses assessing current and new ethical frameworks for autonomous and agentic systems. It also addresses the role of law, regulation and policy in guiding agent development. Finally, it involves philosophical debate around agent consciousness, rights and responsibilities.

4.2 Accuracy, Bias, Fairness, Transparency

Accuracy, bias, fairness, and transparency are core technical and social component. Aspects include methods for improving agent accuracy and self-awareness (e.g., limitations, confidence). This also includes successful approaches to detecting and mitigating dangers such as bias and hallucination in large datasets and models. Techniques for creating more transparent and interpretable models and agents are important as well as evaluating the impact of agents on different demographic groups.

4.3 Human-Agent Collaboration and Trust:

As intelligent agents becomes more integrated into daily life, understanding the dynamics of human-agent collaboration, and the trust between them, is crucial. This includes designing intuitive and trustworthy user interfaces for agentic AI. Another important aspect is the role of agents in decision support systems and associated risks (e.g., awareness, over/under confidence, deskilling). Governance frameworks for autonomous decision-making by AI agents, including responsibility allocation, oversight mechanisms, and human-in-the-loop or human-on-the-loop models are important. Finally, strategies for building public trust and understanding of agent technologies will become important with broad adoption.

4.4 Real-world Applications and Case Studies

As proliferation increases, case studies from various industries will be important to create better governance for responsible development and deployment in low to high risk use cases. Responsible agents will become vital in high assurance applications with human, financial, and security risks. Special attention will be required to address ethical and operational challenges in using agents in high assurance and regulated industries such as healthcare, disaster management, pharmaceutical, financial services, telecommunications, logistics, energy, agriculture, defense, and space. Finaly, AI use in creative fields and its impact on human learning, training, and creativity will be important to understand.

5 Toward a Trustworthy Agents Roadmap

Figure 3 illustrates a roadmap toward safe and secure agents that are worthy of trust. Each of the lanes illustrates a journey over time (ten years) along some particular dimension or route that has various milestones along the way shown in blue. Along the way there are various obstacles (technical gaps) that need to be overcome to advance progress toward agents that are safe and secure and sovereign.

Fig. 3. Roadmap for Safe and Secure Agents.

Progress will be governed by regulation, standards and guidelines. Progress will be limited by the ability to provide appropriate levels of transparency and trust. Agents will progress from general purpose models of today to personalized to muti-domain agents. Agent deployment will advance from individual to multi-agent to ecosystems of agents. Three key lanes of essential progress include confidentiality, integrity and availability, and human-agent interaction and governance.

5.1 Confidentiality

The first lane in the roadmap recognizes confidentiality gaps in areas including privacy leakage, intellectual property theft, model inversion, and copyright violations. Methods for agent identity, secure interagent access, and differential AI privacy are noted in a move toward more controllable agents.

5.2 Integrity and Availability

The second lane of progress addresses gaps in integrity availability including jailbreaking, persuasion, bias, hallucination and model and agent drift. Methods identified including grounding and alignment, explainable AI, bounded execution and formal methods. Future areas such as neurosymbolic models and verifiable and validated models and agents suggest a future with more robust multiagent collaboration.

5.3 Governance and Human-Agent Interaction

The third lane of progress recognizes gaps in governance and human-agent interaction such as the need for agent and inter agent standards (of communication, performance and conduct), runtime controls, methods for delegation of authority and protection of human psychology and cognition. Future areas to more toward more beneficent agents including computational fairness, sustainability, and multiagent conduct and control.

6 Future Directions

Organizations will need to track risks of LLMs closely as law, regulation, recommended practice, technology and threat continue to rapidly evolve. New threats include more sophisticated attacks exploiting the weaknesses of AI systems to undermine their confidentiality, integrity and availability, imposing new harms. Conversely, (human and machine) adversaries can use those AI systems as tools to attack organizations at speed and scale to cause harms. For example, adversaries can masquerade with high fidelity deep fakes to gain access and reveal protected information, corrupt third party information to undermine integrity or reputation, or flood a service with deep fakes to deny access to protected groups.

While this paper offers a roadmap with example mitigations for example legal and regulatory requirements (e.g., privacy and copyright protection), future research by licensed legal/technical experts is required to address legal risks in particular jurisdictions. This could include, for example, comparing national or regional laws (e.g., GDPR Art. 17 vs. US sectoral privacy laws, US Copyright Act § 106A, EU AI Act risk-based obligations) and analyzing how mitigation choices (e.g., data-licensing, watermarking) map into specific statutory or regulatory requirements or exemptions.

Another important category of future risk is the rapidly growing number of AI agents that grow in capability and with increasing levels of delegated authority. Governance across use cases must include specification of agent levels of responsibility and authority and required human awareness and consultation. An important aspect of risk management are controls on agent goals, their access to and use of confidential data, tools, and (financial and human) resources, restricting their ability to make (e.g., high risk) decisions or actions that are inherently human, as well as escalation of high risks to human authorities. In addition, research is required into robust security and resilience controls on Model Context Protocol (MCP) [14] access to other agents as well as tools to discover "shadow AI" [15]. Risk management should anticipate human or agent jailbreaking to subvert limitations (e.g., convincing perhaps unwitting humans to escalate privileges) or outright deception, destruction, or misuse and employ safeguards. New adversarial threats to agentic AI systems target agent APIs, prompt infiltration and obfuscation, credential harvesting, data poisoning and clickbait [16].

As higher level cognitive reasoning and planning develop, research should explore systems that are goal-based which can be infused with a deeper understanding of competence and ignorance, behavioral boundaries, as well as ethical guidelines to increase the likelihood of beneficent behavior. In conclusion, responsible deployment of AI must respect laws, regulations, threat vectors, technical realities, and human limitations to advance human trust and advance the full promise of augmented (human) intelligence

7 Organization of the Collection

This collection is organized around both the theory and practical application of methods to address safety, security, and sovereignty of agents. As illustrated in Table 2, the papers in this collection start addressing the overall threat to and by agents, then move to characterizing and measuring agent safety and security, next to foundational methods for agent trust, and finally turning to methods for effective governance of agents. Accordingly, these four thematic sections address:

1. **Threats** – adversarial availability attacks, prompt-injection environments, and emerging denial-of-service techniques for agents.
2. **Safety, Security and Risk** – neuro-symbolic agents for auditable shop-floor AI, risk modelling for distributed AI, life-cycle risk mitigation for generative models, and the CAR-bench benchmark for LLM-agent consistency.
3. **Trust Building** – cryptographic certificates of AI validity, channel-level safety for agent-to-agent communication, and the human-identity layer for cultural appropriateness.
4. **Governance** – multi-layer governance frameworks, AI-governance policy surveys, infrastructure-first control models, grassroots co-governance, and the open-source ASAGO community for model security.

[The final version of this chapter will highlight some key themes and issues addressed in each of these sections based on the final chapters accepted for inclusion in this book].

Table 2. Collection Organization

Section	Article	Threat	Vulnerability
Threat	ATLAS	Threat Ontology	Across the kill chain
	Agent Threats	Agents	
	AgentDojo	Prompt Injection	LLM vulnerability
	Availability Attacks	Availability	Resource exhaustion
Safety and Security	Industrial Agents	Global manufacturing	Availability and Integrity

	LLM Life Cycle	Across the life cycle	LLM vulnerabilities
Foundations	Crypto Agent Certificates	Agent Identity	No agent identity management
Governance	Frameworks	Immature Regulation	Regulatory Gaps

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7.1 How to Use this Book Target Audience

While all readers may find value in each of the above sections and individual readers may have particular interests in parts of this collection for a variety of purposes, specific sections may be most useful given a particular aim. These include, for example:

- **Graduate students** completing masters and doctorates in AI, robotics, cyber-physical systems, and technology policy should find this collection valuable in both making sense of the vast and highly fragmented literature as well as identifying thesis focus opportunities that address key gaps as well as envisioning methods to the future to advance resilient if not anti-fragile agents.
- **Researchers** in AI safety, formal verification, multi-agent systems, and AI governance will find the sections on safety and security and foundations of agent resilience highly relevant.
- **Commercial and industrial engineers** developing autonomous platforms (automotive, robotics, energy, finance) will find both the introductory section on active threats against agents and their vulnerabilities useful in framing risk, as well as the methods in sections 3 and 4 for mitigating those system threats important.
- **Policy-makers & standards bodies** will find especially section 4 on governance useful in navigating regulations (e.g., EU AI Act), standards (e.g., IEEE 7000) and directives (e.g., DoD 3000.09) both nationally and internationally.

Conclusion

This book aims to provide a unified, multi-disciplinary roadmap which is essential to design, verify, and govern safe, trustworthy, and culturally appropriate autonomous agents. Given the exponentially expanding capabilities of single and multiple agents, the growing dependence on agents across all sectors of the world economy, and the rapidly expanding set of attacks against the confidentiality, integrity and availability of agents, this collection is intended to provide both the focused attention and inspire stronger methods to help secure what is rapidly becoming a critical infrastructure and essential set of agentic services that promise to improve efficiency and effectiveness of many sectors. Given that existing agent safety and security literature is fragmented across conference proceedings, arXiv pre-prints, and disparate standard documents, and that little attention has been paid to sovereign agents, this collection is intended to rapidly disseminate insights into advancing the resilience of agents across multiple domains.

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References

1. Guo, T., Chen, X., Wang, Y., Chang, R., Pei, S., Chawla, N.V., Wiest, O., Zhang, X.: Large Language Model Based Multi-agents: A Survey of Progress and Challenges. In: Proceedings of the 33rd International Joint Conference on Artificial Intelligence (IJCAI 2024), Jeju, South Korea, pp. 8048–8057 (2024
2. International Organization of Standardization and International Electrotechnical Commission (ISO/IEC) TS 5723:2022 Trustworthiness – Vocabulary, https://www.iso.org/standard/81608.html, last accessed 2026/9/7.
3. U. S. Department of Commerce National Institute for Standards and Technology (NIST) Artificial Intelligence Risk Management Framework (AI RMF) 1.0, https://airc.nist.gov/airmf-resources/airmf/3-sec-characteristics, last accessed 2026/9/7.
4. Maybury, M.: Mitigating risks across the life cycle of generative artificial intelligence, Safe Agents Workshop, *Int Joint Conference on AI (IJCAI).* Univ of Bremen, Germany. 16 August 2026.
5. EU AI Act, https://artificialintelligenceact.eu/ai-act-explorer, last accessed 2026/9/7.
6. Maybury, M.: Mitigating biased, brittle and baroque generative AI. In *Proc. of IEEE First International Conference on AI and Data Analytics (ICAD)*, Tufts Univ., Medford, MA, 24 Jun 2025.
7. AI Security Governance Framework 2.0, Cyberspace Administration of China (CAC), September 15, 2025.
8. Autonomy in Weapon Systems. DoD Directive 3000.09. 25 January, 2023.
9. Interim Measures for the Management of Generative Artificial Intelligence Services. Cyberspace Administration of China (CAC), 10 July 2023.
10. Federal Ministry of Education and Research (BMBF) AI Action Plan, Germany, 7 November 2023.
11. America's AI Action Plan, July 2025. White House.
12. *AI Regulation: A Pro-Innovation Approach* (White Paper), HM Government, United Kingdom, 2023.
13. *General Data Protection Regulation* (Regulation (EU) 2016/679), European Union. Official Journal L 119, pp. 188-, 4 May 2016.
14. Parra, D. and Spahr-Summers, J. *Model Context Protocol Specification*. 2025. Anthropic. modelcontextprotocol.io.

15. Scale trusted AI with watsonx.governance. 2026. www.ibm.com/products/watsonx-governance, last accessed 2026/9/7.
16. MITRE ATLAS August 2026 Update. atlas.mitre.org/resources/updates/2026-08, last accessed 2026/9/7.